

Project code

```
pitching <- read.csv("pitching_2015_2024.csv")
```

```
#selecting only pitchers with more than 50 IP because they will most likely have the most consistent data  
pitching_selected <- pitching[pitching$IP > 50, c("WHIP", "ERA", "HR.9", "Age", "K.9", "Team", "Season")]
```

```
model1 <- lm(ERA ~ WHIP, data = pitching_selected)  
summary(model1)
```

```
##  
## Call:  
## lm(formula = ERA ~ WHIP, data = pitching_selected)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.64326 -0.45208  0.00269  0.44100  2.70695   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -2.00352    0.08047  -24.90  <2e-16 ***  
## WHIP         4.71282    0.06258   75.31  <2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.7074 on 3139 degrees of freedom  
## Multiple R-squared:  0.6437, Adjusted R-squared:  0.6436   
## F-statistic: 5671 on 1 and 3139 DF,  p-value: < 2.2e-16
```

```
model2 <- lm(ERA ~ WHIP + HR.9, data = pitching_selected)  
summary(model2)
```

```
##  
## Call:  
## lm(formula = ERA ~ WHIP + HR.9, data = pitching_selected)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.01805 -0.34588 -0.00269  0.33146  2.46692   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -2.05389    0.06273  -32.74  <2e-16 ***  
## WHIP         3.82546    0.05260   72.72  <2e-16 ***  
## HR.9         1.03597    0.02300   45.05  <2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5514 on 3138 degrees of freedom
## Multiple R-squared:  0.7836, Adjusted R-squared:  0.7835
## F-statistic: 5683 on 2 and 3138 DF,  p-value: < 2.2e-16
```

```
anova(model1, model2)
```

```
## Analysis of Variance Table
##
## Model 1: ERA ~ WHIP
## Model 2: ERA ~ WHIP + HR.9
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1    3139 1570.91
## 2    3138  953.97  1    616.95 2029.4 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model3 <- lm(ERA ~ WHIP + HR.9 + K.9, data = pitching_selected)
summary(model3)
```

```
##
## Call:
## lm(formula = ERA ~ WHIP + HR.9 + K.9, data = pitching_selected)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.94523 -0.34570 -0.00743  0.32821  2.51431
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.642426   0.097139  -16.908 < 2e-16 ***
## WHIP         3.709218   0.056419   65.745 < 2e-16 ***
## HR.9         1.028750   0.022926   44.872 < 2e-16 ***
## K.9        -0.029411   0.005319   -5.529 3.48e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5488 on 3137 degrees of freedom
## Multiple R-squared:  0.7857, Adjusted R-squared:  0.7855
## F-statistic: 3834 on 3 and 3137 DF,  p-value: < 2.2e-16
```

```
anova(model2, model3)
```

```
## Analysis of Variance Table
##
## Model 1: ERA ~ WHIP + HR.9
## Model 2: ERA ~ WHIP + HR.9 + K.9
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1    3138  953.97
## 2    3137  944.76  1    9.2074 30.573 3.48e-08 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

model4 <- lm(ERA ~ WHIP + HR.9 + K.9 + Age, data = pitching_selected)
summary(model4)

##
## Call:
## lm(formula = ERA ~ WHIP + HR.9 + K.9 + Age, data = pitching_selected)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.94239 -0.34125 -0.00674  0.32840  2.52122
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.412398   0.127994 -11.035 < 2e-16 ***
## WHIP         3.695075   0.056593  65.293 < 2e-16 ***
## HR.9         1.032245   0.022937  45.003 < 2e-16 ***
## K.9        -0.030077   0.005319  -5.655 1.7e-08 ***
## Age        -0.007351   0.002667  -2.756 0.00589 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5482 on 3136 degrees of freedom
## Multiple R-squared:  0.7862, Adjusted R-squared:  0.786
## F-statistic: 2884 on 4 and 3136 DF, p-value: < 2.2e-16

anova(model3, model4)

## Analysis of Variance Table
##
## Model 1: ERA ~ WHIP + HR.9 + K.9
## Model 2: ERA ~ WHIP + HR.9 + K.9 + Age
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1    3137 944.76
## 2    3136 942.48  1     2.2827 7.5956 0.005885 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

vif(model4)

##      WHIP      HR.9      K.9      Age
## 1.361824 1.170459 1.213753 1.009367

model5 <- lm(ERA ~ WHIP + HR.9 + K.9 + Age + Team, data = pitching_selected)
summary(model5)

##
## Call:
## lm(formula = ERA ~ WHIP + HR.9 + K.9 + Age + Team, data = pitching_selected)
```

```

##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.90244 -0.34040 -0.01323  0.33266  2.51326
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.416307   0.140039 -10.114 < 2e-16 ***
## WHIP         3.663800   0.058349  62.791 < 2e-16 ***
## HR.9         1.032689   0.023161  44.587 < 2e-16 ***
## K.9        -0.026143   0.005422  -4.822 1.49e-06 ***
## Age        -0.006634   0.002759  -2.405  0.01624 *
## TeamARI     -0.026835   0.063439  -0.423  0.67232
## TeamATL     -0.095313   0.065257  -1.461  0.14423
## TeamBAL      0.010242   0.063054   0.162  0.87098
## TeamBOS     -0.026569   0.061419  -0.433  0.66534
## TeamCHC     -0.166509   0.065074  -2.559  0.01055 *
## TeamCHW     -0.022786   0.065509  -0.348  0.72799
## TeamCIN     -0.066977   0.064111  -1.045  0.29624
## TeamCLE     -0.059861   0.060661  -0.987  0.32381
## TeamCOL      0.181370   0.062464   2.904  0.00372 **
## TeamDET      0.090252   0.061193   1.475  0.14035
## TeamHOU     -0.045996   0.061565  -0.747  0.45505
## TeamKCR      0.020813   0.064026   0.325  0.74515
## TeamLAA     -0.043743   0.061960  -0.706  0.48025
## TeamLAD     -0.085015   0.064710  -1.314  0.18902
## TeamMIA      0.041890   0.062576   0.669  0.50327
## TeamMIL     -0.134823   0.061707  -2.185  0.02897 *
## TeamMIN      0.075015   0.063853   1.175  0.24016
## TeamNYM      0.023376   0.064483   0.363  0.71699
## TeamNYY     -0.067943   0.063982  -1.062  0.28836
## TeamOAK      0.150337   0.063936   2.351  0.01877 *
## TeamPHI     -0.014946   0.064183  -0.233  0.81589
## TeamPIT     -0.064058   0.065213  -0.982  0.32603
## TeamSDP      0.014664   0.063768   0.230  0.81814
## TeamSEA      0.023355   0.066711   0.350  0.72629
## TeamSFG      0.015750   0.061232   0.257  0.79702
## TeamSTL     -0.068370   0.065032  -1.051  0.29319
## TeamTBR      0.014778   0.066569   0.222  0.82433
## TeamTEX      0.058152   0.066173   0.879  0.37959
## TeamTOR     -0.053241   0.068515  -0.777  0.43718
## TeamWSN     -0.104343   0.065953  -1.582  0.11373
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5463 on 3106 degrees of freedom
## Multiple R-squared:  0.7898, Adjusted R-squared:  0.7875
## F-statistic: 343.2 on 34 and 3106 DF, p-value: < 2.2e-16

```

```
anova(model4, model5)
```

```

## Analysis of Variance Table
##
## Model 1: ERA ~ WHIP + HR.9 + K.9 + Age

```

```
## Model 2: ERA ~ WHIP + HR.9 + K.9 + Age + Team
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    3136 942.48
## 2    3106 926.83 30      15.65 1.7483 0.007169 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(model4)
```

```
##      WHIP      HR.9      K.9      Age
## 1.361824 1.170459 1.213753 1.009367
```

```
pitching_selected$COL <- ifelse(pitching_selected$Team == "COL", 1, 0)
pitching_selected$CHC <- ifelse(pitching_selected$Team == "CHC", 1, 0)
pitching_selected$MIL <- ifelse(pitching_selected$Team == "MIL", 1, 0)
pitching_selected$OAK <- ifelse(pitching_selected$Team == "OAK", 1, 0)
```

```
modelTeam <- lm(ERA ~ WHIP + HR.9 + K.9 + Age + COL + CHC + MIL + OAK,
               data = pitching_selected)
summary(modelTeam)
```

```
##
## Call:
## lm(formula = ERA ~ WHIP + HR.9 + K.9 + Age + COL + CHC + MIL +
##      OAK, data = pitching_selected)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.93253 -0.34340 -0.00707  0.33689  2.52739
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.421381   0.127768 -11.125 < 2e-16 ***
## WHIP         3.671732   0.056677  64.784 < 2e-16 ***
## HR.9         1.032040   0.022844  45.177 < 2e-16 ***
## K.9         -0.028266   0.005317  -5.316 1.13e-07 ***
## Age         -0.006638   0.002661  -2.495 0.01266 *
## COL          0.192340   0.056947   3.377 0.00074 ***
## CHC         -0.151792   0.059887  -2.535 0.01131 *
## MIL         -0.119609   0.055899  -2.140 0.03245 *
## OAK          0.162808   0.058330   2.791 0.00528 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5459 on 3132 degrees of freedom
## Multiple R-squared:  0.7883, Adjusted R-squared:  0.7878
## F-statistic: 1458 on 8 and 3132 DF, p-value: < 2.2e-16
```

```
anova(model4, modelTeam)
```

```
## Analysis of Variance Table
```

```
##
## Model 1: ERA ~ WHIP + HR.9 + K.9 + Age
## Model 2: ERA ~ WHIP + HR.9 + K.9 + Age + COL + CHC + MIL + OAK
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    3136 942.48
## 2    3132 933.28   4    9.1912 7.7112 3.519e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model6 <- feols(ERA ~ WHIP + HR.9 + K.9 + Age + COL + CHC + MIL + OAK | Season, data = pitching_selected)
summary(model6)
```

```
## OLS estimation, Dep. Var.: ERA
## Observations: 3,141
## Fixed-effects: Season: 10
## Standard-errors: IID
##           Estimate Std. Error  t value    Pr(>|t|)
## WHIP    3.677402    0.056808 64.73347 < 2.2e-16 ***
## HR.9     1.025568    0.023443 43.74766 < 2.2e-16 ***
## K.9     -0.031559    0.005442 -5.79922 7.3279e-09 ***
## Age     -0.007548    0.002664 -2.83307 4.6400e-03 **
## COL      0.190481    0.056839  3.35123 8.1410e-04 ***
## CHC     -0.150833    0.059784 -2.52298 1.1686e-02 *
## MIL     -0.120037    0.055776 -2.15213 3.1463e-02 *
## OAK      0.160636    0.058244  2.75797 5.8500e-03 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.543007      Adj. R2: 0.788809
##                               Within R2: 0.785632
```

```
p_ols <- plm(ERA ~ WHIP + HR.9 + K.9 + Age + COL + CHC + MIL + OAK, data = pitching_selected, model = "fe")
p_fe <- plm(ERA ~ WHIP + HR.9 + K.9 + Age + COL + CHC + MIL + OAK, data = pitching_selected, model = "fe")
phtest(p_fe, p_ols)
```

```
##
## Hausman Test
##
## data: ERA ~ WHIP + HR.9 + K.9 + Age + COL + CHC + MIL + OAK
## chisq = 24.297, df = 8, p-value = 0.002043
## alternative hypothesis: one model is inconsistent
```

```
model7 <- lm(ERA ~ WHIP + HR.9 + K.9 + Age + BB.9 + AVG, data = pitching_selected)
summary(model6)
```

```
## OLS estimation, Dep. Var.: ERA
## Observations: 3,141
## Fixed-effects: Season: 10
## Standard-errors: IID
##           Estimate Std. Error  t value    Pr(>|t|)
## WHIP    3.677402    0.056808 64.73347 < 2.2e-16 ***
## HR.9     1.025568    0.023443 43.74766 < 2.2e-16 ***
```

```
## K.9 -0.031559 0.005442 -5.79922 7.3279e-09 ***
## Age -0.007548 0.002664 -2.83307 4.6400e-03 **
## COL 0.190481 0.056839 3.35123 8.1410e-04 ***
## CHC -0.150833 0.059784 -2.52298 1.1686e-02 *
## MIL -0.120037 0.055776 -2.15213 3.1463e-02 *
## OAK 0.160636 0.058244 2.75797 5.8500e-03 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.543007 Adj. R2: 0.788809
## Within R2: 0.785632
```

```
anova(model4, model7)
```

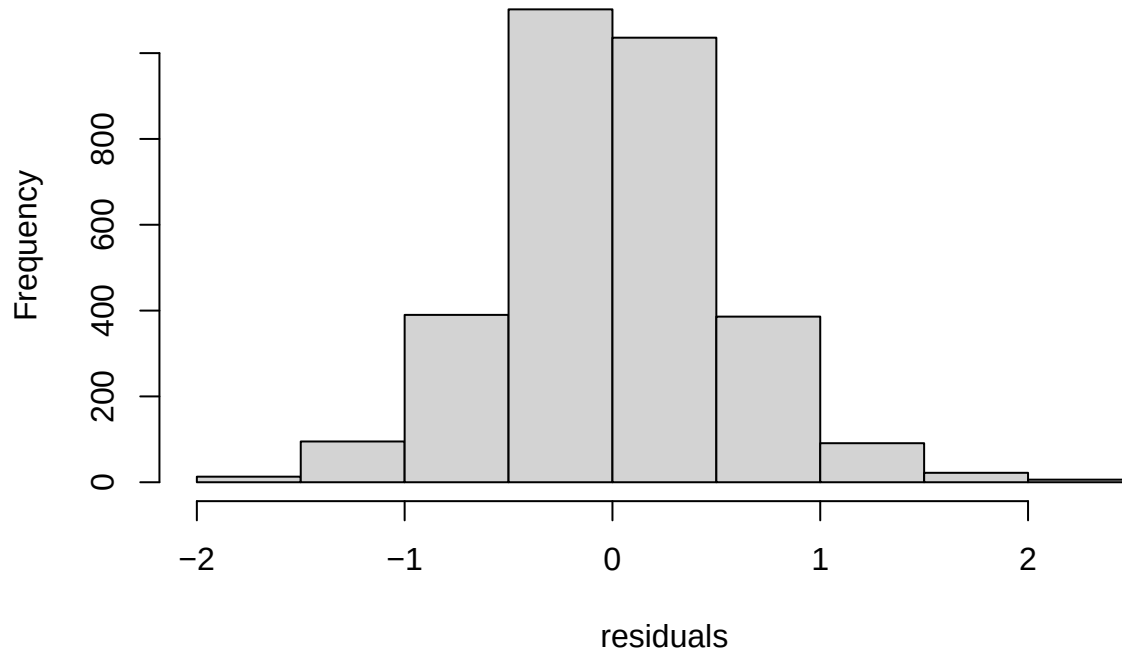
```
## Analysis of Variance Table
##
## Model 1: ERA ~ WHIP + HR.9 + K.9 + Age
## Model 2: ERA ~ WHIP + HR.9 + K.9 + Age + BB.9 + AVG
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1    3136 942.48
## 2    3134 879.91  2    62.569 111.43 < 2.2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(model7)
```

```
##      WHIP      HR.9      K.9      Age      BB.9      AVG
## 144.342276  1.268922  1.770340  1.012805  43.881251 103.329146
```

```
residuals <- residuals(model6)
hist(residuals)
```

Histogram of residuals

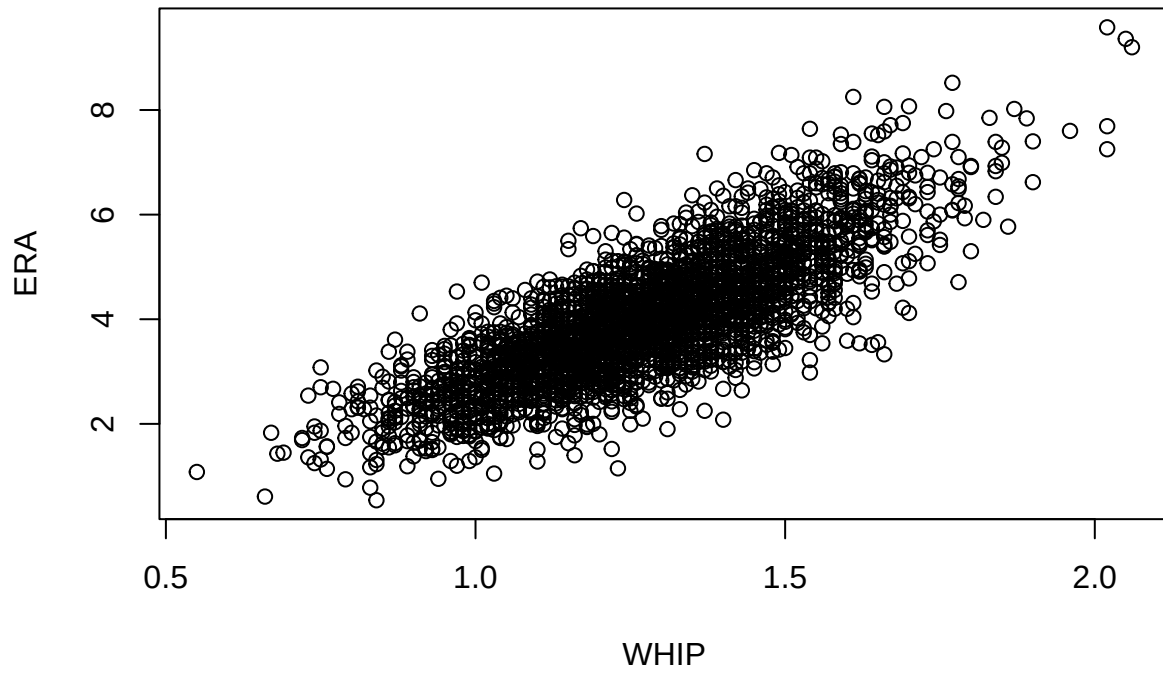


```
#Anderson-Darling Test  
ad_test <- ad.test(residuals)  
ad_test
```

```
##  
## Anderson-Darling normality test  
##  
## data: residuals  
## A = 3.8137, p-value = 1.665e-09
```

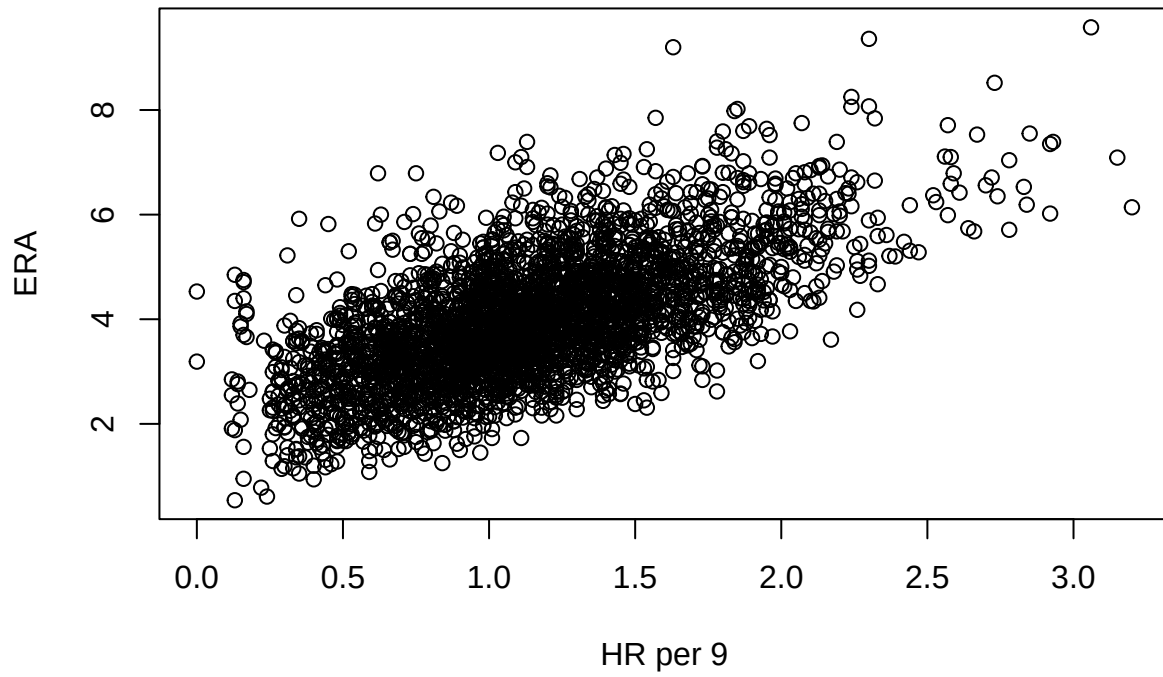
```
plot(pitching_selected$WHIP, pitching_selected$ERA, main = "Relationship Between WHIP and ERA", xlab = "W
```

Relationship Between WHIP and ERA



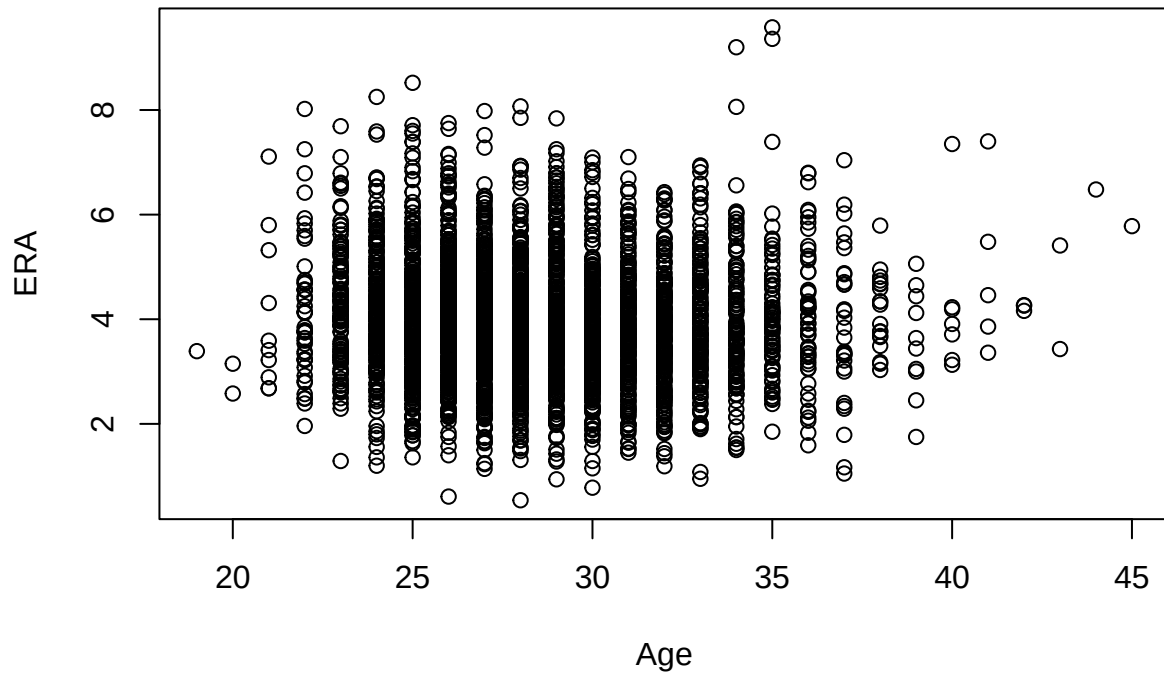
```
plot(pitching_selected$HR.9, pitching_selected$ERA,main = "Relationship Between HRs per 9 and ERA",xlab
```

Relationship Between HRs per 9 and ERA



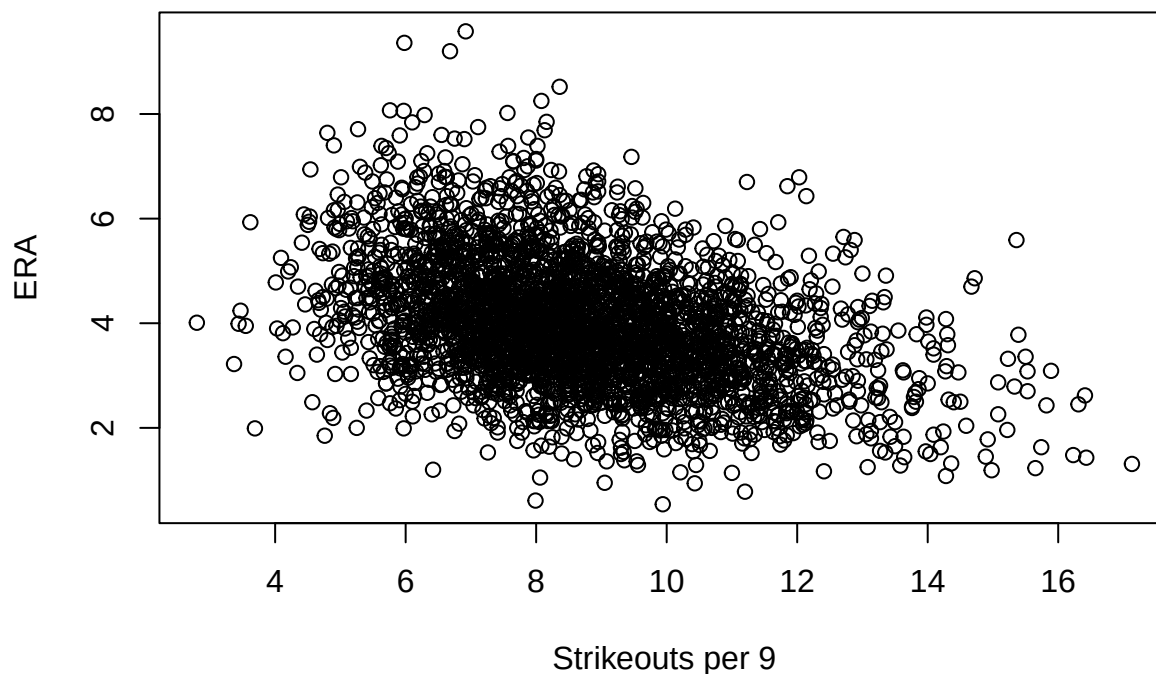
```
plot(pitching_selected$Age, pitching_selected$ERA, main = "Relationship Between Age and ERA", xlab = "Age
```

Relationship Between Age and ERA



```
plot(pitching_selected$K.9, pitching_selected$ERA, main = "Relationship Between Strikeouts per 9 and ERA")
```

Relationship Between Strikeouts per 9 and ERA



```
.007/mean(pitching_selected$ERA)
```

```
## [1] 0.001758078
```

```
# final model removed age because of practical significance
```

```
modelFinal <- feols(ERA ~ WHIP + HR.9 + K.9 + COL + CHC + MIL + OAK | Season, data = pitching_selected)
summary(modelFinal)
```

```
## OLS estimation, Dep. Var.: ERA
```

```
## Observations: 3,141
```

```
## Fixed-effects: Season: 10
```

```
## Standard-errors: IID
```

	Estimate	Std. Error	t value	Pr(> t)
## WHIP	3.691340	0.056659	65.15065	< 2.2e-16 ***
## HR.9	1.022454	0.023443	43.61381	< 2.2e-16 ***
## K.9	-0.030662	0.005439	-5.63755	1.8784e-08 ***
## COL	0.194912	0.056881	3.42665	6.1898e-04 ***
## CHC	-0.159742	0.059768	-2.67268	7.5638e-03 **
## MIL	-0.120015	0.055838	-2.14933	3.1685e-02 *
## OAK	0.162749	0.058305	2.79133	5.2811e-03 **

```
## ---
```

```
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## RMSE: 0.543705 Adj. R2: 0.788334
```

```
## Within R2: 0.785081
```

```

modelTest <- feols(ERA ~ 1 | Season, data = pitching_selected)
pitching_selected <- pitching_selected %>% mutate(ERA_resid2 = resid(modelTest))

# How much of 'educ' is absorbed by FE? ( $R^2 \sim 1 \Rightarrow$  little identifying variation left)
pitching_selected$ERA_resid2 <- resid(modelTest)

era_var2 <- var(pitching_selected$ERA, na.rm = TRUE)
era_residvar2 <- var(pitching_selected$ERA_resid2, na.rm = TRUE)

# "FE R Squared" for the regressor: how much of educ is explained by FE?
r2_fe_era <- 1 - era_residvar2 / era_var2
r2_fe_era

```

```
## [1] 0.02015268
```

```

summary_table <- pitching_selected %>%
  select(-Team, -Season) %>%      # drop non-numeric grouping variables
  summarise(across(
    .cols = everything(),
    .fns = list(
      mean = mean,
      min = min,
      max = max,
      median = median,
      sd = sd
    ),
    na.rm = TRUE
  ))

```

```

## Warning: There was 1 warning in 'summarise()'.
## i In argument: 'across(...)'.
## Caused by warning:
## ! The '...' argument of 'across()' is deprecated as of dplyr 1.1.0.
## Supply arguments directly to '.fns' through an anonymous function instead.
##
## # Previously
## across(a:b, mean, na.rm = TRUE)
##
## # Now
## across(a:b, \(x) mean(x, na.rm = TRUE))

```

```
summary_table
```

```

##   WHIP_mean WHIP_min WHIP_max WHIP_median   WHIP_sd ERA_mean ERA_min ERA_max
## 1  1.269968    0.55    2.06      1.27 0.2017359 3.981621    0.54    9.58
##   ERA_median ERA_sd HR.9_mean HR.9_min HR.9_max HR.9_median HR.9_sd
## 1      3.89 1.184993  1.136434      0      3.2      1.1 0.4614473
##   Age_mean Age_min Age_max Age_median   Age_sd K.9_mean K.9_min K.9_max
## 1 28.60331    19     45      28 3.685214 8.692305    2.8    17.13
##   K.9_median K.9_sd BB.9_mean BB.9_min BB.9_max BB.9_median BB.9_sd AVG_mean
## 1      8.52 2.026344  3.116129      0.6    8.25      3.03 1.00435 0.2395664
##   AVG_min AVG_max AVG_median   AVG_sd COL_mean COL_min COL_max COL_median

```

```

## 1  0.111  0.349      0.241 0.03445017 0.03088188      0      1      0
##      COL_sd  CHC_mean CHC_min CHC_max CHC_median      CHC_sd  MIL_mean MIL_min
## 1 0.1730252 0.02737982      0      1      0 0.1632135 0.03151862      0
##      MIL_max MIL_median      MIL_sd  OAK_mean OAK_min OAK_max OAK_median      OAK_sd
## 1      1      0 0.1747424 0.02897167      0      1      0 0.1677536
##      ERA_resid2_mean ERA_resid2_min ERA_resid2_max ERA_resid2_median ERA_resid2_sd
## 1      1.171841e-16      -3.446135      5.444855      -0.08177966      1.172992

```